

7. Is it right to waste helium on party balloons?



Helium is a colourless inert gas found in Group 0 of the Periodic Table.

Helium is one of the commonest elements in the Universe, second only to hydrogen. However, on Earth it is relatively rare, as shown in **Table 1**.

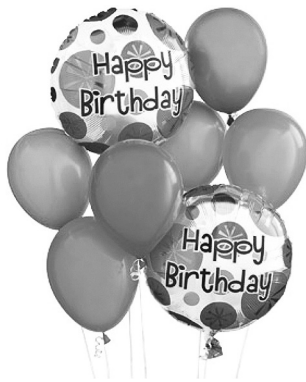
Gases which have a density less than air can escape the Earth’s gravity and leak away into space. The density of air is 1.2g/m³. **Bar chart 1** shows the densities of Group 0 gases.

Helium has the lowest boiling point of any element. This makes it of key importance for magnets used in hospital MRI scanners, which must be super-cooled to generate the hugely powerful magnetic fields required.

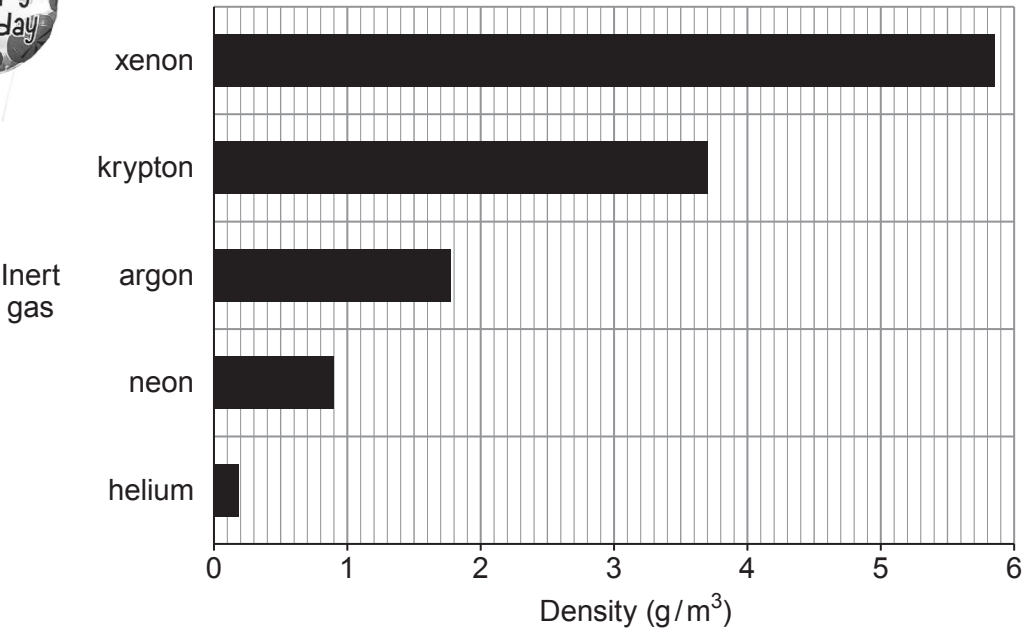
Some scientists believe that because helium is a finite resource it should not be used for party balloons.

Table 1

Inert gas	Percentage in the atmosphere (%)	Melting point (°C)	Boiling point (°C)
helium	0.00052	-272	-269
neon	0.0018	-246	-246
argon	0.93	-186	-186
krypton	0.0001	-152	-152
xenon	0.000009	-111	-106



Bar chart 1



(a) Answer the following questions using the information given.

- (i) Tick (✓) the box next to the **most** important property that makes helium a suitable material to fill **floating** party balloons. [1]

Helium is a gas

☐

Helium is the second most common element in the Universe

☐

Helium is less dense than air

☐

Helium is colourless

☐

- (ii) Tick (✓) the box next to the correct statement. [1]

The Earth's atmosphere contains more helium than argon

☐

The Earth's atmosphere contains more xenon than helium

☐

The Earth's atmosphere contains more helium than krypton

☐

- (iii) Tick (✓) the box next to the **best** reason for not using helium to fill party balloons. [1]

There isn't much helium in the Earth's atmosphere

☐

Scientists say helium shouldn't be used to fill balloons

☐

Helium is a finite resource

☐

- (iv) Tick (✓) the box next to the correct statement. [1]

Only helium gas can leak away into space

☐

Helium and neon gases can leak away into space

☐

Only argon can leak away into space

☐

All inert gases can leak away into space

☐

(b) The table below shows the electronic structure of three Group 0 elements.

Group 0 element	Electronic structure
helium	2
neon	2,8
argon	2,8,8

Tick (✓) the box next to the statement that **best** explains why Group 0 elements are unreactive.

[1]

All Group 0 elements have 2 electrons in their inner shell

☐

All Group 0 elements have 8 electrons in their outer shell

☐

All Group 0 elements have full outer shells

☐

All Group 0 elements have some full shells

☐

5

